

EE147 Assignment #4 Questions

How many global memory reads are performed during the histogram kernel?

- 1) Each element will be read from global memory once by the kernel. Since we are reading from array N, the amount of global reads that will be performed is the size of N.

How many global memory writes are performed during the histogram kernel?

- 2) The final histogram has 16 blocks and 1024 bins. This means each block contains a private histogram with 1024 bins. Once the block is finished processing its portion of the input, it will write all 1024 bins into global memory. As a result, we get 16×1024 or 16384 total global memory writes.

How many atomic operations are performed in shared memory?

- 3) Each input element increments a bin in the block's private histogram. Looking at our while loop, we are staying in the loop until i is greater than the size of the array. Therefore, each element is processed once. We process them into shared memory the same amount of times as size of N. We perform N amount of atomic operations in shared memory.

How many atomic operations are performed in global memory?

- 4) Each block has a private histogram with 1024 bins. During the merge step, each block adds its 1024 private bins into the global histogram using atomic add. Since the while loop will perform 16 iterations for the 16 blocks, and each iteration performs 1024 atomic adds. We have 16×1024 or 16384 total atomic adds performed in global memory.

Now assume we have a naive implementation of reduction (without privatization).

How many global memory atomic operations are performed in total?

- 5) In naive implementation, each input element directly increments a bin in the global histogram. Since there are N input elements and each element performs one atomic operation, the total number of global memory atomic operations is N.